

Project Scope of Work

Project Name: CollabHub (Chat + Task Management Platform)

1. Project Overview

CollabHub is a lightweight collaboration platform designed for teams to communicate and manage tasks within shared channels.

The platform combines:

- Real-time communication (chat)
- Channel-based collaboration
- Task management within channels

Each channel acts as a workspace where team members can:

- Communicate via chat
 - Create and manage tasks
 - Assign responsibilities
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2. Project Objectives

The primary goal of this application is to:

- Enable seamless team communication
 - Provide basic task tracking within channels
 - Ensure real-time updates for collaboration
 - Maintain a simple and scalable architecture
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3. User Roles

3.1 Standard User

- Can register and log in

- Can join channels
- Can send/receive messages
- Can create and manage tasks within channels

(Role-based permissions are not fully implemented yet)

4. Functional Requirements

4.1 Authentication

- User registration
 - User login
 - JWT-based authentication
 - Session persistence (token-based)
 - SSO
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4.2 Channel Management

- Create a new channel
 - Join an existing channel
 - View list of channels
 - View channel members
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4.3 Real-Time Chat

- Send messages within a channel
 - Receive messages in real-time
 - Maintain message history (basic)
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4.4 Task Management (Within Channels)

Each channel supports task operations:

- Create a task
- Assign task to a channel member
- Update task status:

- To Do
 - In Progress
 - Done
 - View list of tasks per channel
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4.5 Real-Time Updates (Expected Behavior)

- New messages should appear instantly
 - Task updates should reflect in real-time
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5. Technical Stack

Frontend

- React (Vite)
- Tailwind CSS
- Material UI

Backend

- Node.js
- Express.js

Database

- MongoDB (Live connection)

Real-Time Communication

- Socket.io
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6. Non-Functional Requirements

- Basic performance optimization
- Maintainable code structure
- Scalability considerations (future-ready)
- Error handling and logging (basic)

7. Deliverables (Expected System Behavior)

- Functional authentication flow
- Working channel system
- Real-time chat functionality
- Task management within channels
- Basic UI for all features

8. Pending / Future Enhancements (Not Fully Implemented)

- Role-based access control (admin/member)
- Notifications (task updates, mentions)
- File sharing in chat
- Task activity logs
- Advanced filtering/search
- Scalable socket handling (Redis/pub-sub)

9. Known Limitations (Current System)

- Limited validation
- Inconsistent API responses
- Partial real-time implementation
- Basic UI without optimization
- No strict architectural separation

10. Partially Completed Work

The following features are already developed but may be incomplete, buggy, or not aligned with best practices:

10.1 Authentication

- Basic signup/login implemented
- JWT-based auth exists

- Missing validation and error handling
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10.2 Channel Management

- Channel creation and joining available
 - Channel member structure exists
 - Access control not properly enforced
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10.3 Real-Time Chat

- Socket connection established
 - Message sending/receiving implemented
 - Issues in:
 - Message duplication
 - Inconsistent delivery
 - Improper socket handling
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10.4 Task Management

- Task creation implemented
- Task assignment available
- Task status updates exist

Issues in:

- Status updates not reflecting properly
 - Assignment inconsistencies
 - API structure not standardized
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10.5 Task Listing UI

- UI components partially built
 - Data not properly synced with backend
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10.6 Channel Members

- Backend support exists
 - Frontend integration incomplete
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10.7 Real-Time Task Updates

- Socket events partially implemented
 - UI not fully reacting to updates
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11. Summary

CollabHub is a partially completed collaboration platform that integrates chat and task management.

While core functionalities exist, the system requires:

- Debugging
 - Refactoring
 - Completion of incomplete features
 - Improved structure and scalability planning
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